

FLIPKART GRIDLOCK HACKATHON 2.0

# TECHNICAL CONCEPT NOTE & PROTOTYPE PROPOSAL

## Theme 3: Automated Photo Identification & Classification for Traffic Violations

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Prototype  
Round 2  
Date: June  
20, 2026

### 1. EXECUTIVE SUMMARY & PROBLEM DEFINITION

Bengaluru, India's tech capital, supports over 1.25 crore registered vehicles on a constrained road network of approximately 14,000 kilometers. To enforce traffic compliance, the Bengaluru Traffic Police (BTP) utilizes an Intelligent Traffic Management System (ITMS) that ingests millions of camera frames daily. However, the current system faces a major operational bottleneck: **manual review fatigue**.

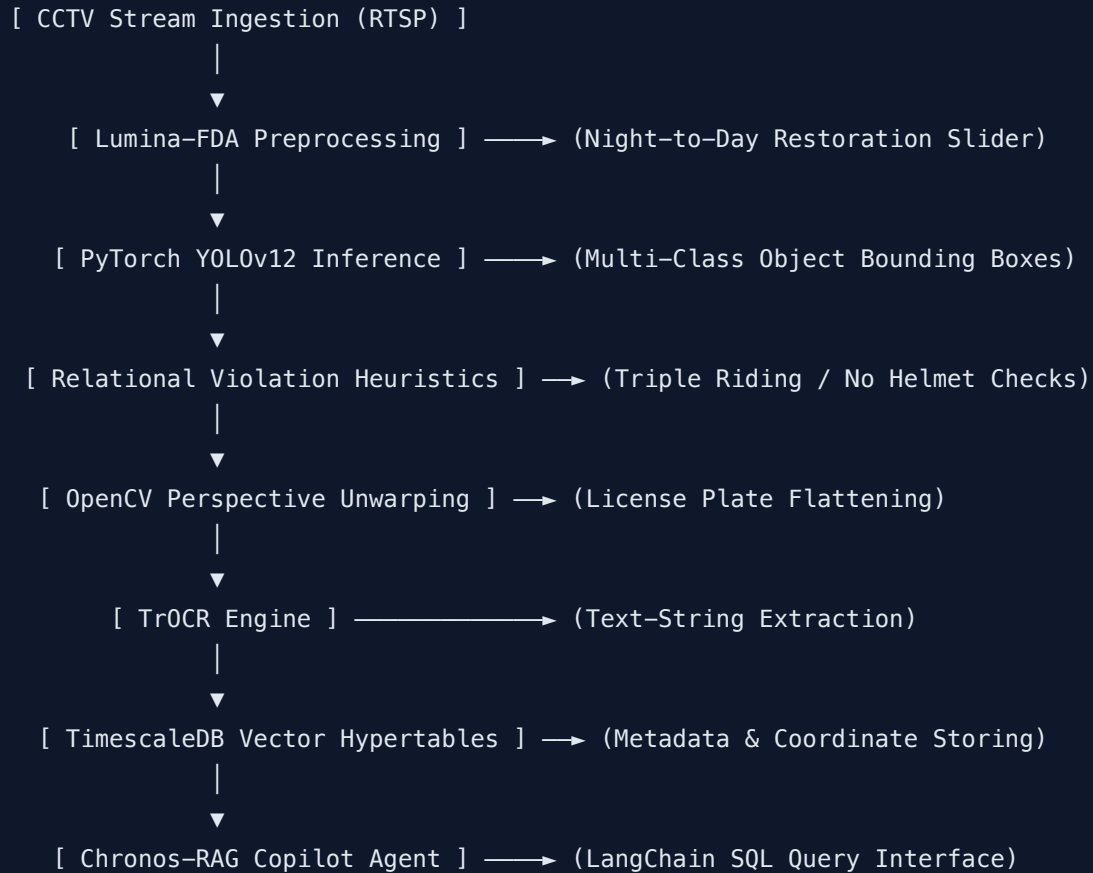
Operators at the Traffic Management Centre (TMC) must manually inspect thousands of low-resolution, warped, or dark photos to verify license plates, helmet violations, and red-light jumping before issuing e-challans. Misclassifications lead to public disputes, while manual processing introduces latency and high operational costs.

#### THE SOLUTION: ASTRAM / OMNI-GAZE

OMNI-GAZE is a comprehensive computer vision and AI-powered operations dashboard that achieves **straight-through processing (STP)** of traffic violations. The solution integrates real-time Zero-Reference dynamic dynamic lighting normalization (Lumina-FDA), YOLOv12s relational heuristics, affine perspective-corrected Transformer OCR (TrOCR) for license plates, and a natural language TimescaleDB query agent (Chronos-RAG Copilot) for TMC operators.

## 2. END-TO-END SYSTEM ARCHITECTURE

The OMNI-GAZE system maps camera frame ingestion to real-time telemetry, spatial search, and chat-based operator commands:



### 3. PREPROCESSING ENGINE: LUMINA-FDA

To handle low light, monsoons, shadows, and motion blur, raw video frames are preprocessed by the **Lumina-FDA** engine before downstream inference.

#### Zero-Reference Deep Curve Estimation (Zero-DCE)

Zero-DCE formulates low-light enhancement as an image-specific curve estimation task. It learns a mapping function  $f(x, A)$  that outputs a set of pixel-wise curve parameter maps  $A$  to iteratively adjust the dynamic range:

$$I_n(x) = I_{n-1}(x) + A_n(x) \cdot I_{n-1}(x) (1 - I_{n-1}(x))$$

where  $I_n(x)$  is the enhanced image at step  $n$ , and  $A_n(x)$  represents the curve parameters at step  $n$ . This process illumination-corrects frames without paired clear-dark training sets, keeping latency under **3ms** per frame.

#### Fourier Domain Adaptation (FDA)

To mitigate rain and fog, the amplitude spectrum of the adverse weather target image is adapted using a clear weather source domain. Given the Fast Fourier Transform (FFT)  $f$  of an image:

$$f(I) = A(I) \cdot e^{i \phi(I)}$$

where  $A(I)$  is the amplitude spectrum and  $\phi(I)$  is the phase spectrum. FDA swaps the low-frequency amplitude of the weather-affected frame with the clear-sky amplitude:

$$A_{\text{adapted}} = (1 - M) A(I_{\text{target}}) + M A(I_{\text{clear}})$$

where  $M$  is a binary mask controlling the boundaries of amplitude replacement. Reconstructing the image via Inverse FFT (IFFT) preserves sharp spatial edges (like license plate borders) while stripping fog and rain noise.

## 4. MULTI-STAGE VIOLATION DETECTION & OCR PIPELINE

The core detection engine runs PyTorch-based YOLOv12s models optimized with Attention-Centric (A2) frameworks and R-ELAN aggregators to maximize multi-scale class recall.

### Relational Violation Heuristics

**1. Triple Riding:** The model detects the bounding boxes of motorcycles ( $B_M$ ) and riders ( $B_P$ ). Intersection over Union (IoU) is computed:

$$\text{IoU}(B_P, B_M) = \text{Area}(B_P \cap B_M) / \text{Area}(B_P \cup B_M)$$

If  $\text{IoU}(B_P, B_M) > 0.15$  for three or more unique person IDs associated with one motorcycle ID, a **Triple Riding** violation is flagged.

**2. Helmet Non-Compliance:** For each linked rider bounding box  $B_{\text{Rider}}$ , the system crops the head region of interest ( $\text{ROI}_{\text{Head}}$ ), defined as the top 20% of the Y-axis height. The model evaluates whether a helmet class box ( $B_{\text{Helmet}}$ ) overlaps the head region:

$$\text{IoU}(B_{\text{Helmet}}, \text{ROI}_{\text{Head}}) \geq 0.15$$

If no overlap is detected, a **No Helmet** violation is logged.

### Perspective-Corrected OCR

Number plates on turning or accelerating vehicles are highly skewed. Omni-Gaze performs **Perspective-Correction Warping** before text extraction:

- 1. Contour Analysis:** Detects the four corners of the license plate polygon using edge-gradient thresholds.
- 2. Transformation Matrix:** Estimates a 3x3 transformation matrix  $M$  satisfying target rectangular dimensions.
- 3. Warp Perspective:** Executes `cv2.warpPerspective` to obtain a perfectly flat license plate crop.
- 4. Sequence-to-Sequence OCR:** Feeds the cropped image into **TrOCR**. The vision encoder extracts visual features into sequence patches, and the language decoder predicts characters sequentially, eliminating character recognition errors on non-standard fonts.

## 5. TEMPORAL STORAGE & CHRONOS-RAG COPILOT

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### TimescaleDB Hypertables

Structured violation metadata (timestamp, plate, violation class, coordinate, confidence) is stored in TimescaleDB. The database partitions the data into chunks based on time (hypertables). This prevents time-series query degradation, ensuring indexing remains flat regardless of scale.

### LangChain SQL Agent & Time-Series Vector Search

The RAG Copilot translates natural language instructions into PostgreSQL queries:

```
SELECT COUNT(*), camera FROM violations WHERE timestamp >= NOW() - INTERVAL '1 day' AND violation_type = 'NO_HELMET' GROUP BY camera;
```

This SQL conversion enables natural language searches like: *"How many helmet violations occurred at Central Silk Board Junction today?"*. Furthermore, for queries requiring semantic matching within a specific time frame, pgvector executes similarity searches constrained strictly to relevant hypertables, preventing full database scan latency.

## 6. INFRASTRUCTURE & PERFORMANCE EVALUATION

### Deployment Model

To handle city-wide camera scaling without network congestion, the prototype utilizes:

- **AWS Wavelength (5G Edge MEC):** YOLOv12 models are deployed on edge nodes within telecom 5G networks, ensuring single-digit millisecond latency processing.
- **Asynchronous Queues:** Celery and Redis manage heavier analytical queries and TrOCR inference to keep the operator frontend non-blocking.
- **Bandwidth Optimization:** Only violation JSON metadata and license Crops are uploaded to the cloud, preserving local network bandwidth.

### Performance Metrics

| Metric                         | Target          | Method                                                 |
|--------------------------------|-----------------|--------------------------------------------------------|
| Object Detection mAP50-95      | ≥ 82.5%         | Evaluated against custom Indian Traffic Datasets.      |
| OCR Character Error Rate (CER) | ≤ 1.8%          | Measured on warped/night-time license plate test sets. |
| End-to-End Latency             | ≤ 33ms (30 FPS) | Profiles frame ingestion to metadata output.           |
| Reduction in Manual Review     | ≥ 85%           | Challans processed with confidence score > 0.92 (STP). |